Special Section: Smart Agriculture

Designing a Smart Honey Supply Chain for Sustainable Development

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***Abstract*—Smart agricultural systems can ensure food security at the source, combat counterfeits and mislabeled goods, garner and preserve trust, and reduce the price of product differentiation. In the honey sector, a smart supply chain could help lift beekeepers out of poverty by generating better market opportunities and value for their honey and strengthening apiculture's position as a vital component of sustainable agriculture. In doing so, the proposed system improves food security and food safety, and reduces honey fraud while strengthening greater biodiversity. This article shows how an open, image-based traceability system designed for sustainable development can be built and used around the world. Key smart agricultural technologies discussed include data-driven honey yield prediction and verification, a smart distribution system built on blockchain technology, pollen signature verification with machine learning algorithms, and a final customer information portal. Taken together, this smart agricultural system has the potential to further several of the United Nations' Sustainable Development Goals.**

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Health risks from adulterated and counterfeit foods is a common consumer health issue.[1] Indeed, incidents of food fraud or economically motivated adulteration (EMA) have been systematically defined as public health concerns.[2] As

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incidents of food fraud increase, van Ruth[3] defines opportunities, motivations, and control measures as the three main elements of food fraud vulnerability. Often food is willfully mislabeled for economic gain which damages health, safety, and trust.[1] Smart agriculture technologies improve food supply chain security by aiding verification of quality at the source, thus enhancing food safety and consumer confidence and knowledge.[4] The prospect of developing an economical and portable system for food safety, which ensures quality from production to the consumer, has been called a game changer[1] for the food industry.

Because so much of our food supply also depends on pollination by managed honey bees, creating a smart traceability system for apiculture would enhance the viability and rewards for beekeepers and have a broad positive impact on our food system thus supporting The United Nations' (U.N.) Sustainable Development Goals (SDGs). These SDGs were born with the idea of having a positive impact on the world based on 17 key areas.[5] First among these goals is to eliminate poverty. While there is no one solution to poverty, a smart honey supply chain can enhance sustainable development by supporting a more economically viable apiculture industry. There can also be benefits for sustainable crop production agriculture where pollination services are needed.

Beekeeping has been described as an ideal, accessible, and empowering opportunity for rural microentrepreneurs in economically challenged areas.[6] Comparatively low labor requirements and start-up costs in combination with minimal land use are just some of beekeeping's competitive advantages for on-farm integration and off-farm deployment, particularly for people with limited access to natural and financial resources.[7,8,9] On a global level, beekeeping produces vital ecosystem services by pollinating 75% of the types of crops grown,[10] and by maintaining phytobiodiversity.[11]

However, the main consumer product from beekeeping, honey, is one of the most adulterated high value foods in the world.[12] This issue has an adverse effect on the viability of apiculture by limiting the premium honey market to a select few trusted sources, locking many in the Global South out of the differentiated market.[12] As a result, most smallholder producers are further confined to their local market or bulk sales abroad through intermediaries competing on price, thereby reducing smallholder income.[13]

Enhancing the ability of smallholder farmers to reach markets and actively engage in them is one of the most pressing economic development challenges.[14,15] For example, research on similar products, such as fair trade coffee, shows that consumers are willing to pay an average 10% price premium, while supporters of the fair trade program are willing to pay up to 27% more.[16] Moreover, Arnot *et al.*[17] find that purchasers of fair trade-labeled coffee are less price-sensitive compared to their peers. Further research shows that many consumers are willing to pay up to 21% more for the geographic labeling.[18]

Research by Cazier *et al.*[19] shows that aligning with consumers' values (such as fair trade) and enhancing trust lifts both purchase intentions and price premiums independently. This approach increases both sales volume and marginal profits, thereby providing more economic benefits for the Global South. A smart honey traceability system, adapted to the needs and technology available in the Global South, can help unlock these profitable markets to beekeepers by building consumers' trust through a honey verification system and by leveraging shared values when beekeepers communicate their story, thus contributing to a reduction in poverty while supporting several related SDGs.

Openness is an essential feature in a smart supply chain for development. It makes it easier for farmers to use with their current beekeeping process. Openness also increases the growth rate of the data needed for the system to become "smart" through machine learning technologies and to be accepted by a large audience of consumers, thus making it useful. These two features of *perceived ease of use* and *usefulness* have been at the heart of the technology acceptance model since 1989[20] and are essential to this system's adoption, having been validated many times over the last 30 years.[21]

We propose a smart honey traceability system based on smart phone images: this system would be open and easy-to-use for the Global South and provide better access to markets and

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higher prices. Smart phone images are ideal for the Global South because they

a) are location aware;
b) are language agnostic;
c) are accessible to all education levels;
d) are harder to fake consistently without detection than simply recording text information;
e) contain metadata that can be used for verification and validation efforts;[22] and
f) are now relatively affordable and common even in the Global South,[23] which takes advantage of an existing resource.

The proposed smart honey traceability system makes the following *major contributions* to society and the literature.

- *Reduce Honey Fraud:* Honey is one of the most faked foods in the world. This system would reduce that fraud through smart tracking of the production and supply chain and smart verification of authenticity with the pollen signature.
- *Increase Food Safety:* Fraudulent Honey poses a risk to consumers' health as well as the economic loss. This system can trace food to the source, not just verify it is real, increase food safety and producer accountability.
- *Reduce Poverty:* This smart system can help reduce poverty by giving beekeepers better access to premium markets. It also gives a path for jobless and disadvantaged groups to participate in the economy more efficiently due to the low barriers to entry of beekeeping compared to other agriculture markets that require access to land to grow crops.
- *Crop Production and Biodiversity:* By making beekeeping more attractive economically, farmers will benefit from better pollination of crops, increasing yield and nature will benefit through more effective pollination improving natural growth and development.
- *Available Technology:* This smart supply chain solution shows how to use current proven technology available and adapted to the developing world in a novel and smart way to have a real impact on the world. For a solution to work in challenging areas, it is critical to build it on technology they can afford, understand, and use, thus increasing its impact and success.
- *Advanced Technology:* While the data collection system uses proven available technology, the analysis system on the data collected is state of the art with image processing, blockchain technology, pollen signature verification, and algorithmic backtracking providing a unique and novel solution to a real problem.
- *Example:* This system can serve as a model for other systems for smart agricultural projects to traceability.

TENETS OF A TRACEABILITY SYSTEM

A traceability system that starts collecting data during production has the advantage of proving honey is real at the outset rather than trying to identify adulterated honey at the end of the supply chain. This paradigm shift requires proving integrity in production (volume and type), validating distribution (chain of custody), and product verification at the final sale (pollen signature). Each of these pieces require smart agricultural systems for production, distribution, and a smart consumer interface to verify the product history and authenticity.

We show how accessible communication systems with verified users and a smartphone camera can provide the data needed via text messages, WhatsApp, WeChat, Signal, or a dedicated application. These methods are preferred to email or web submissions because they are location-aware (on the smartphone) and users are verified by the provider, giving more accountability and reducing gamesmanship.

The data needed for this system include images and occasional responsive messages from the system through the beekeepers preferred communication tool. The images needed vary depending on whether beekeepers are selling generic honey or premium honey based on origin, variety, or management practices (e.g., sustainable practices) as illustrated in Figure 1.

In the next sections, we show how the collected data, recorded on a blockchain and supplemented with secondary data, machine learning, and statistical algorithms, can provide

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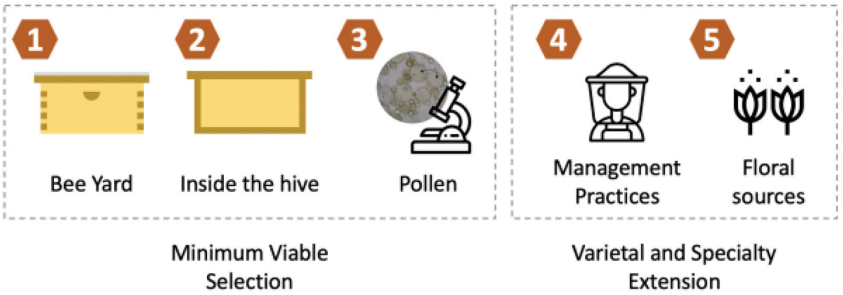

Figure 1. Core and extended data points.

verifiable production and distribution and a traceable pollen signature.

DATA COLLECTION AND INTEGRITY

To register, the beekeeper sends a message to the system from a verified communication platform. In a quick responsive back and forth, their account information is recorded along with information about the camera (metadata), number of hives and yards they have, and the type of beekeeper they are (smallholder stationary, migratory, queen breeder, etc.). Also, they are asked if their honey has any premium attributes such as varietal or raw, or minimal processing (such as filtering and heating), and whether their beehives receive specific management techniques (e.g., least-toxic pest management, organic). These responses can be written in any major language (detected based on location and original text). If needed, an NGO or government organization can provide one-time registration help. Once registered, beekeepers submit images of

a) the apiary and surrounding landscape;
b) the inside of the hives;
c) the pollen;
d) the floral food source;
e) water sources;
f) potential risk factors (e.g., sources of toxins, competition); and
g) any management practices they perform each time they work an apiary.

Most are normal images taken in and around the apiary.

Typically, submission rates would be 2–3 times a month during the season.

The pollen image, however, is taken with an attachable clip-on microscope that attaches to the smartphone camera, such as the one in Figure 2. Such a microscope is ideally suited as it is available for ~$35 USD and requires no external power, slides, or wires, and provides magnification up to 400×. Images can be taken by the beekeeper when they visit the field or during harvest at a facility. Several images need to be taken from different hives in the apiary to get a good sample distribution for analysis.

He *et al.*[25] demonstrated microscopic pollen image analysis could be a robust tool for honey authentication. He *et al.*[26] used 320× images to build a machine learning algorithm that was able to identify pollen families from comparable images. With the addition of bloom images, a *Pollen Signature* can then be checked for varietal source consistency.

All the images are validated and standardized through image processing to ensure consistency with the location, and images from prior visits and the metadata can be checked to make sure the camera, location, and time the picture was taken is consistent with local norms for those activities. Once verified, the beekeepers' data are written to the system database. A one-way hash of the data are then generated and added to a system-wide daily batch hash, generated with a Merkle Tree Algorithm and written to a public facing blockchain platform, like Ethereum, to ensure data integrity.

A one-way hash transforms any amount of data by applying a cryptographic algorithm, resulting in a fixed-length code.[27] Anyone with the original data and the correct algorithm would get the same hash, and it is impossible to reverse and generate the same data from the hash.[27] This process allows the hash to serve as an immutable snapshot of the data for a given point in time while protecting privacy.

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Figure 2. Example of a ~$35 attachable smartphone microscope [24].

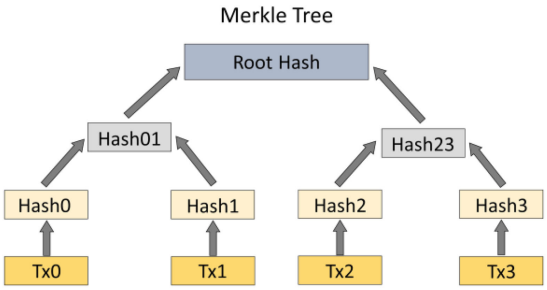

Figure 3. Merkle Tree.

A Merkle Tree is a way of combining multiple fixed-length hashes together in a tree-like structure (two at a time) with each generating a new fixed-length hash from two prior hashes as shown in Figure 3. This feature is commonly used for blockchain integrity[28,29] and is easily verified with a fast rehash of the data.[30] In this way, only the daily root hash needs to be written to a public blockchain to verify data integrity, with intermediate hashes recorded in the system database for auditability.

When it comes time to harvest the honey, the beekeeper submits a request for a harvest voucher via their registered messaging system. In the request, they report how many hives they harvested per yard and the total volume of honey produced in each yard. They also note any special features, such as "this yard was sourwood honey" or "this yard is organic" to be verified. They next submit a number of additional images of the honey harvest process and, in particular, the pollen signatures of that yard. There needs to be enough images to give a valid statistical pollen distribution for that yard for future verification.

VOLUME VERIFICATION

The system uses the data to verify that the amount and type of honey reported by the beekeeper is *plausible* given the collected data. This process includes checking the volume produced, varietal reported, and management actions taken.

The volume verification is based primarily on the location, plants of the area, time stamps, and internal hive images collected throughout the season. With location information obtained from the metadata of images, actual volumes can be compared to a statistical range of estimated volumes. These volume checks include making sure a) the harvest volume per hive is consistent with historical averages for this area; b) the harvest volume is consistent with what is reported by other beekeepers in similar phytoclimatic regions that same year; and that c) the harvest volume is consistent with algorithmic projections of the honey that could have been produced given known conditions on the ground, including proxies for hive health.

The first two volume checks are self-explanatory, but the third deserves some additional description. The first and second algorithmic projections are based on the *Flight* and *Cluster* hours the bees would have experienced during the season based on the locations, weather, and daylight. Bees only fly to gather nectar when weather conditions are right. These conditions include a temperature between 55° (12.8 °C) and 100 °F (37.8 °C), winds less than 15 MPH (24.1 KM/H), no rain, and adequate daylight.[31]

Given the location data from relevant images, sunrise and sunset can be calculated.[32] Then, the nearest public weather station's data are pulled and analyzed, and finally the flight hours the bees would have experienced can be calculated. The images from inside the hive can be used to estimate hive strength and honey storage space vectors over time (limited storage space will affect yield, and too much space may impact the ability of the colony to defend from pests and may signify the presence of diseases). With hive strength, space, and flight hour information, an estimate of the amount of honey produced can verify yield.

This process can be enhanced with data from hive scales or bee counters to refine the

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algorithm with daily honey production scores based on weight and adjusted for human impact.[33] Furthermore, the flight hours of prior years, which can be computed retroactively, can be used with historical records to adjust the estimates up or down. As the system matures, it can be further refined with penalties for *Cluster Hours* when the temperature falls below 50 °F (10 °C) and the bees spend excess energy (honey consumption and "shivering") thermoregulating cluster temperature so the whole colony can survive low external temperatures.[31] It can also be adjusted downward for bloom availability due to adverse weather conditions, drought, hive density, or other factors that reduce the food supply. Landscape images of the yard as well as satellite and land use data can also be used to assess apiary suitability.

VARIETAL AND PRACTICE VERIFICATION

The varietal and specialty honeys require extra work with *Algorithmic Backtracing*. For the varietal, the system would start with the bloom images and use artificial intelligence algorithms to do image matching to the claimed floral type at each time period and check that verified images were taken at the right locations and times to produce the varietal. With *Transfer Learning,* a neural network could be built on currently available labeled data from plants. By adapting that model with images of bee-relevant blooms (varietals), a new model could be built with fewer labeled images.[34] Incorporating existing data from *ImageNet* or the *World Bee Count* would make this approach more practical. Next it would use distribution analysis on pollen signature images to make sure the pollen in the honey matches bloom pollen in the pollen database, within statistical normal ranges for that varietal.

Management practices are harder to verify with images and would benefit from third-party verification and auditing. However, there are still several ways to benefit beekeepers with these images, including practice verification. First, consider that any management practice is defined by both actions taken and those not taken. Someone might attempt to claim a label like "organic" by choosing *not* to take an image of an inconsistent treatment to game the system. However, a label such as organic or treatment-free is not just the absence of things you do not want. Those bees still need to be managed. If anything, it requires more work with different techniques. Those additional practices would generate additional images of work *not* done by conventional beekeepers, but which are required by the chosen method. It is the absence of these required images that the system could be trained to look for. Faking these images at the same location and with the same hives would likely be cost-prohibitive to someone trying to cheat the system, even if they were familiar with those methods, though many will not be if they do not practice them. Additionally, even the time distribution of pictures taken (or not) and other characteristics for those attempting fraud would be features that could be used to train a model.

While this process can be undertaken with current image processing and labeled data, a new machine learning method, called *Zero-Shot Learning,* may be able to accelerate the building of the algorithm. Zero-shot learning self-generates labels to analyze image features based on large bodies of text.[34] With a large enough corpus of text about identified bee practices, the algorithm can be trained to look for images consistent with those practices, even without labeling. In addition, select images or other data of management practices can be shared with the final purchaser increasing trust, connection, and value congruence to increase purchase intention and willingness to pay price premiums.[19] Finally, the images allow for the possibility of a human expert audit based on the images, creating another deterrent and increasing trust.

CHAIN OF CUSTODY

Much of the value of a traceability system is in opening up higher value markets by increasing value congruence and trust[19] while reducing the cost of product differentiation. Establishing a chain of custody that ensures the honey was not adulterated after leaving the apiary is critical for increasing that trust.

To address this issue, a transfer log could be generated for each sale or even a transfer within

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a larger organization. The seller (or mover) would send (via the app, Whatsapp, WeChat, Signal, or text, etc.) the buyer's ID (or shipping location) along with the yard ID(s) and the range of units sold (i.e., yard ABCD123 units 5 through 555) to record that transfer. These units could be deducted from the old ledger and moved to the new with a hash of the transfer data being added to the daily root hash. Since all units are tracked, this process could reduce the risk of a double sale. Sample images of the pollen could be used to create an audit trail.

POLLEN SIGNATURE

One of the keys to this system is the *Pollen Signature Check,* which was inspired by work from He *et al.*[25,26] referenced earlier. The algorithms would scan sample images for pollen families to verify they are congruent with the origin images. If consistent, a positive match would be returned by the system. This signature check serves as a verification system where it looks for a match to a known signature based on pollen ratios, not just presence or absence. In addition, this process may potentially pick up on other key features (i.e., color, moisture) that may be in the honey to help with verification. This process is different from a search system looking for a match with many possible solutions; it is more akin to the face validation found on a smartphone, using *one-shot learning*.[34] For example, a Siamese Neural Network (SNN) can be trained on one image using this technique.[34] Because this is a known search architecture focused on verification, it is much less computationally intensive than a hard target search where it tries to identify from a large database of options. *Data Augmentation* could also be used to add minor transformations (i.e., rotate, flip, darken, etc.) to labeled images and synthetically expand the dataset for analysis.[34,35,36] A clustering algorithm to match each honey sample with new data could further improve the verification process.[34] These approaches would also help with floral recognition.

The final buyer, and all those in between, could take a sample of images of the honey with a microscope and verify that the key signature is present. Once confirmed, the system could open up a honey storyboard. In the storyboard, select images, location, and other information about the beekeeper and the trail of honey from the beekeeper to the buyer could be displayed, building a deeper connection between the buyer and producer. Figure 4 gives an overview of the system in practice.

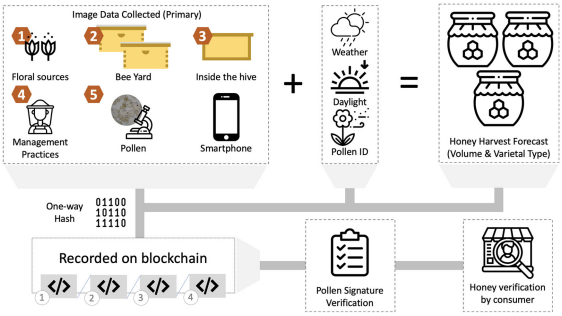

Figure 4. System overview.

IMPLEMENTATION DETAILS

In terms of implementation, recall that the system is made up of three interrelated parts: 1) production verification; 2) distribution tracking; and 3) final customer portal, which includes the pollen signature check and production information. It is therefore likely that the system would be rolled out in stages: first, with production verification, next with chain of custody, and finally with the pollen signature check and information portal. For this practical reason, organizations such as *Maya Bees* and *Bees for Development* that control most of their own supply chains would be ideal candidates for testing an early version of this system as the chain of custody is mostly already established and consumers buy in part for value congruence, which independently increases trust.[19] For those without this tight supply chain or value congruence, the entire system might be needed to get the full benefits.

Additionally, by partnering with organizations like those listed above that already have boots on the ground and established networks, the system could be rolled out much faster by integrating into existing networks and production systems. Because this system combines existing, proven technology for data

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gathering in a novel way to develop a smart supply chain, the rollout of the data gathering phase should happen relatively quickly. It would take time and quality data, however, to fully develop the algorithms for verification of the honey production. A partner organization could also help by ensuring quality data collection and on-the-ground verification. The distribution system could be built now with existing blockchain technology. The part of the consumer information that would display information about the product could be built quickly; however, the pollen signature matching would require time and data to build.

LIMITATIONS AND FUTURE RESEARCH

Even with an image-based system, some training may be required for certain types of beekeepers, but this could be added on to existing training programs, making it more viable.

The system does start to break down when bulk honey is blended with other varieties as it would lose some of its resolution for a pollen signature check for unique traceability. This situation could be compensated for with images from each yard being blended with samples of the final blend to show statistical plausibility of the new pollen ratios based on the original inputs.

With enough data, a natural extension could be the development of a unary classification machine learning model that would allow the system to check for authenticity of samples never seen before. These systems have the advantage of not needing labeled data (i.e., fraud or not fraud), but look instead at a large sample of data, checking for outliers, anomalies, or novel findings. Common algorithms include one-class Gaussian, one-class K-Means, unary kNN, and one-class SVM.[34]

Over time, with enough pollen grain information, it is likely that the pollen matching algorithms and databases would become rapid and robust enough to provide a simpler form of honey authentication based just on location and honey sampling during production, which would reduce the need for other verification checks in the system and allow for just the assignment of a production ID for the yard. This is another reason to support an open system that can accelerate data acquisition globally and offer additional services to help economic development. Until such systems are developed and proven, some occasional assistance from local partners or labs to provide additional audit capability may be needed.

While microscope use for magnification discussed in the paper (400×) has been shown to detect pollen families[25,26] the practicality of using the microscope in the field still needs testing. However, even without that feature the system could go a long way toward providing trust to consumers. Additionally, He *et al.*[25,26] demonstrated that a low cost highschool grade microscope was able to take high enough quality pictures to be useful, providing another possibility for these to be used at the packing facility on regular samples taken from the field if needed.

CONCLUSION

A smart agricultural system focused on honey traceability is a viable solution to the problem of economically motivated honey adulteration taking place and is well adapted to be used in areas of low literacy, language ambiguity, and limited access to expensive technologies, exactly the kind of places where an increasing amount of our global food supply is being produced. The system has the potential to be a "gamechanger" as described by Masana,[1] for not just honey, but for farmers dependent on a healthy supply of beekeepers to ensure adequate pollination. If proven with beekeepers, this type of system could be used as a template for bottom up certification of other agricultural products.

This novel and practical solution also has a high potential to be used for rural economic development, especially in the Global South where it can help open new markets for honey to those most in need. We encourage NGOs, governments, and other parties to support the development and adoption of this system.

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